

FINAL PITCH

TEAM NAME: Gangs Of KHARAGPUR

COLLEGE: IIT Kharagpur

PROBLEM STATEMENT:

Traditional EHR systems are centralized, easy to breach, and limit patient control. Our blockchain-based secure EHR system (Vital Block) ensures data integrity, patient ownership, and seamless credential verification while enabling privacy through selective disclosure and secure interoperability in healthcare

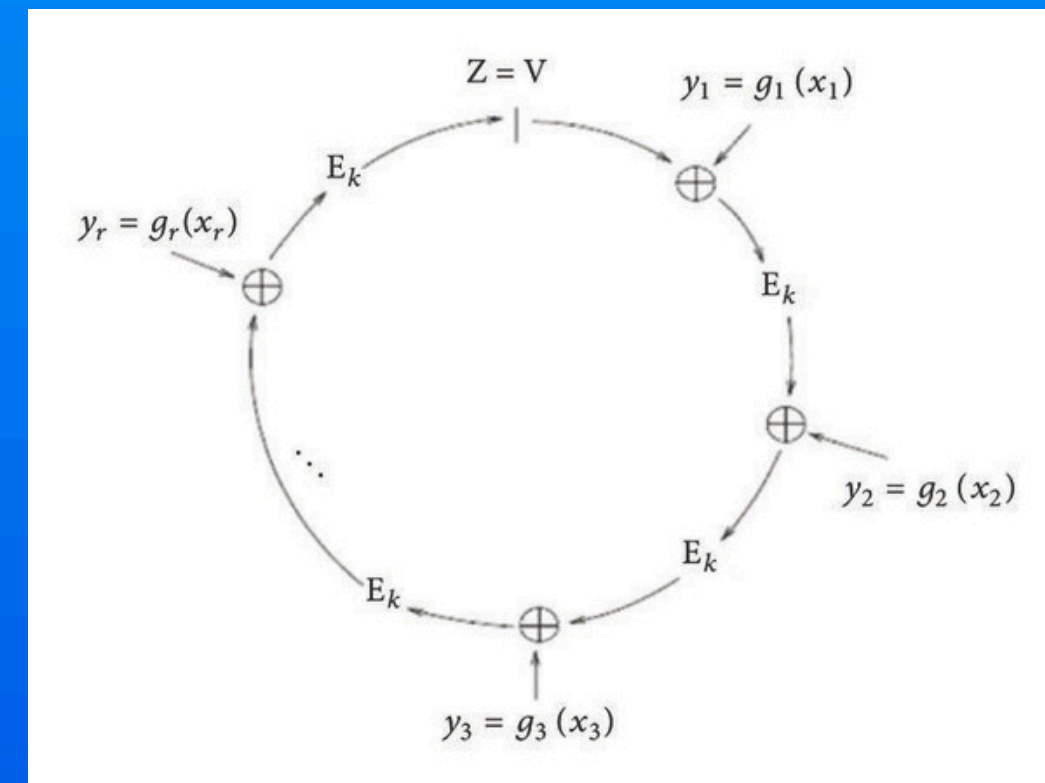
PROPOSED SOLUTION

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We have worked on a niche related to use of Decentralized Identity in the health-care industry. The blockchain-based hospital system ensures secure, tamper-proof, and decentralized storage of patient records, transactions, and other critical hospital operations. The solution leverages Ethereum blockchain with Zero-Knowledge Proofs (ZKPs) to maintain patient privacy while ensuring data integrity. Further for verification of certificates we have thought about implementing a signature method somewhat akin to Ring Signatures which will allow someone to reveal that they have a certificate knowing that the only way the other party will be able to reveal that fact to third parties only by revealing their private key.



Ring Signature Algorithm



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SOLUTION ARCHITECTURE

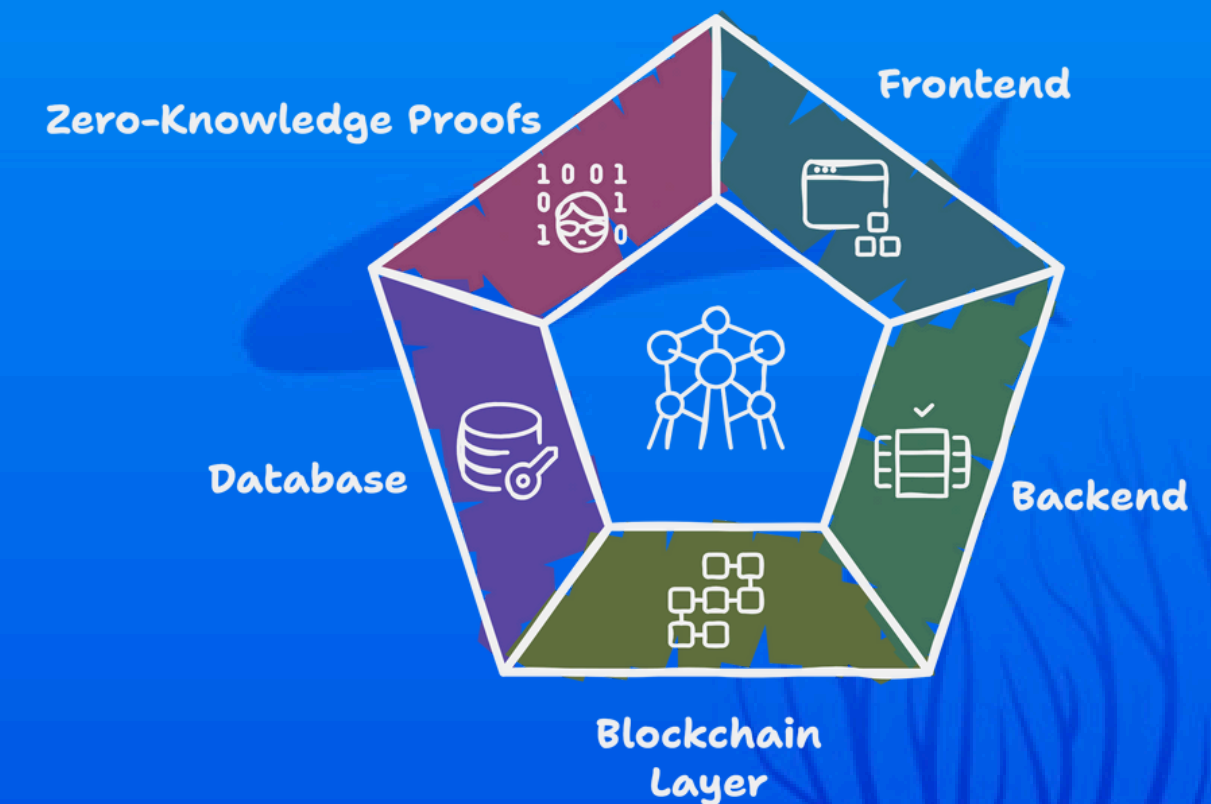
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The system consists of the following key components

- **Frontend:** A user-friendly web interface built using Java Script and Tailwind.
- **Backend:** A blockchain-powered backend using Node.js (Express.js) and Smart Contracts on Ethereum.
- **Blockchain Layer:** Ethereum (Layer 1) or for on-chain transactions and security.
- **Database (Off-Chain Storage):** IPFS (Inter Planetary File System) for encrypted off-chain data storage.
- **Zero-Knowledge Proofs (ZKPs):** Utilized for verifying sensitive medical data without exposing it



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UNIQUE SELLING PROPOSITION

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1. Privacy-Preserving Data Sharing with Zero-Knowledge Proofs (ZKPs)

- ◆ Hospitals and research institutions can verify patient information without accessing actual data.

2. Smart Contract-Based Appointments & Billing

- ◆ Automated appointment scheduling, medical billing, and insurance claim settlements through smart contracts—reducing paperwork and fraud.

3. Decentralized Doctor & Staff Identity Verification

- ◆ Healthcare professionals' credentials are stored on-chain, ensuring that only verified doctors can access sensitive data.

4. Fraud Prevention in Insurance & Prescriptions

- ◆ Digitally signed prescriptions and verified insurance claims prevent fake prescriptions and fraudulent claims.



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DEMO

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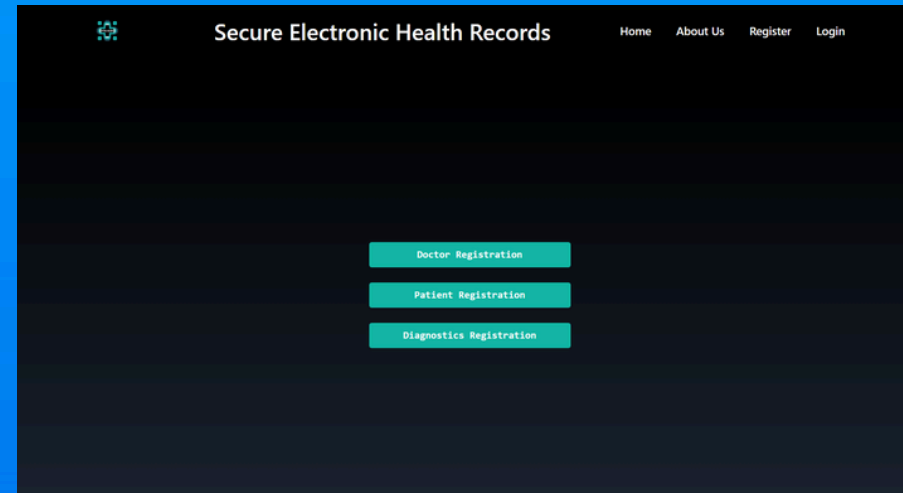
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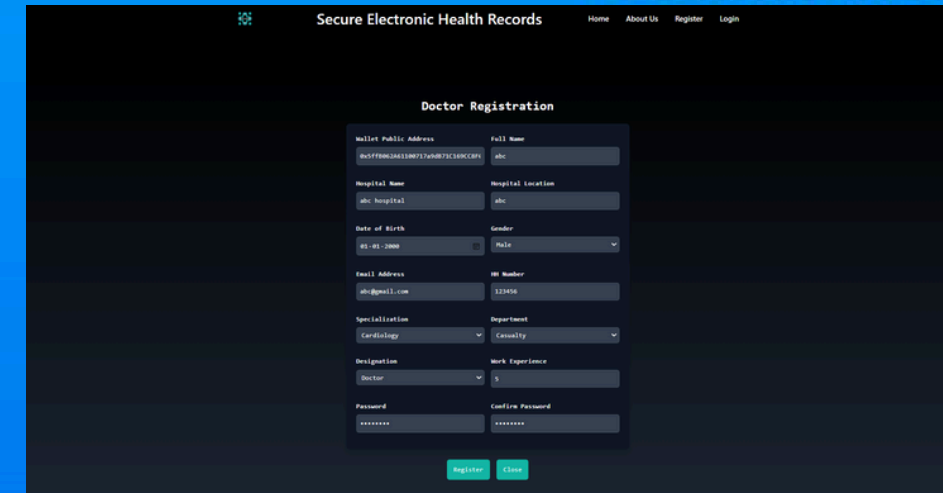
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Interface – connect Metamask



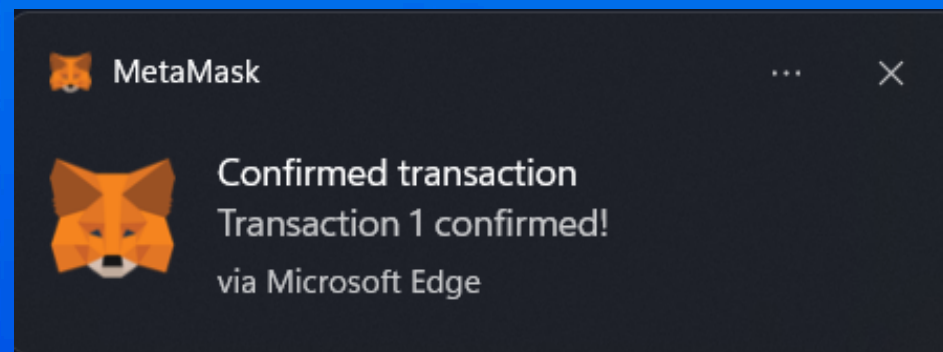
Registeation Interface



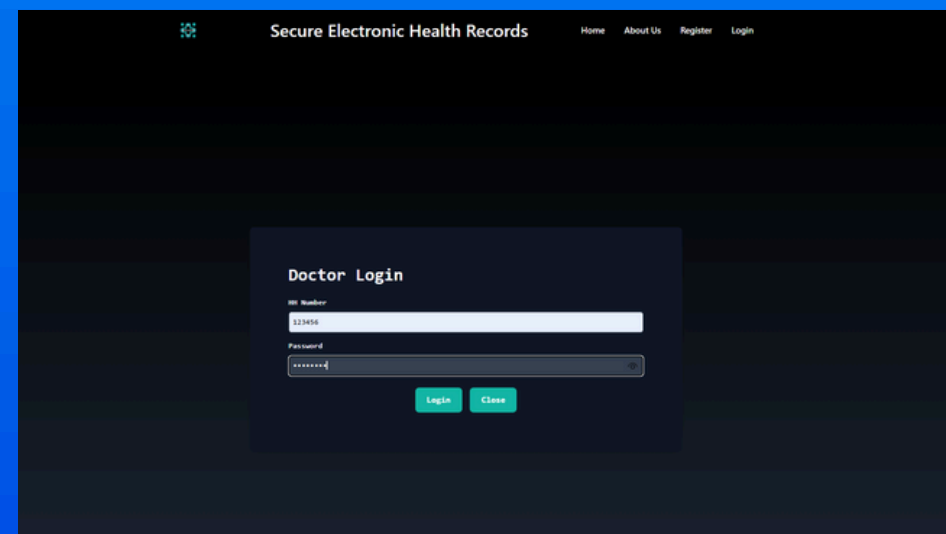
Doctor Registration



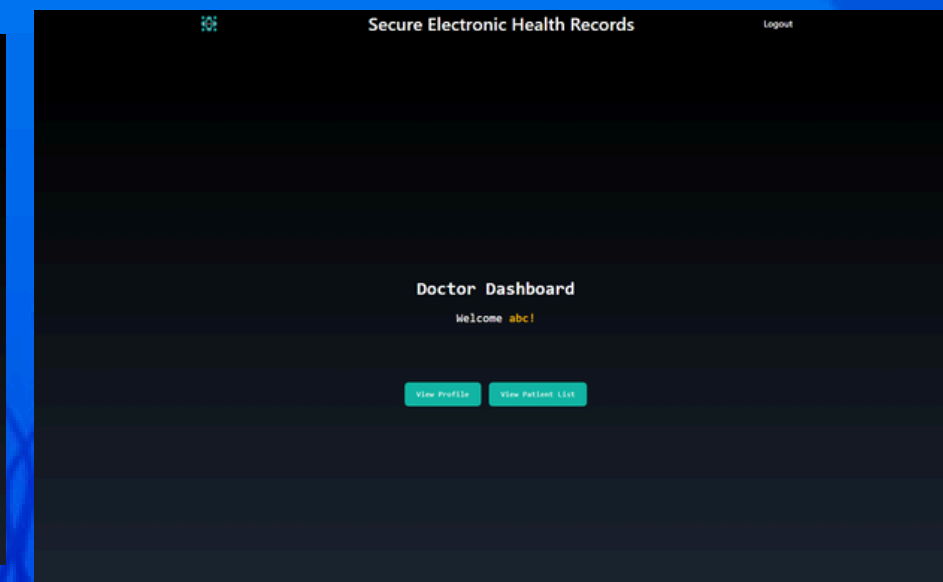
On register it will send a request to Ganache – ethereum testnet



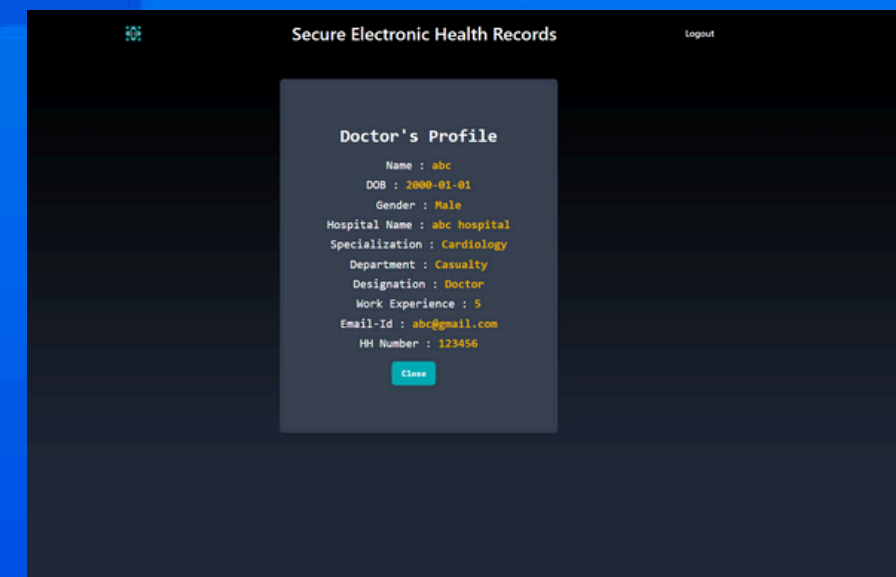
After successfull transaction,
a message will be printed



Login Interface



You have successfully created
your Decentral-ID



You can access your
information



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BUSINESS MODEL

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CODE TO INFINITE POSSIBILITIES
LARGEST HEALTHCARE INNOVATION CHALLENGE

Revenue Streams



Subscription Fees

In order to gain access, healthcare providers must pay a regular price.



Transaction Fees

A cost associated with requests for consent or data sharing occurrences.



Premium Services

Offer healthcare providers analytics and forecasting information.



Licensing

Provide technology and APIs to external parties (e.g., insurers)



Development costs

Building and maintaining the blockchain platform



Infrastructure

Hosting blockchain nodes and storage systems



Compliance and Legal

Following with laws pertaining to healthcare, such as GDPR or HIPAA.



Customer Support

Providing 24/7 assistance for platform users

Cost Structure

Go-to Market Strategy



Partnerships

Collaborations with hospitals, clinics, and EHR providers



Marketing

Emphasize patient empowerment and operational efficiency



Pilot Programs

Start in a few chosen institutions to show their efficacy.



Scaling Up

Following early success, reach out to businesses and academic institutions.



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TARGET MARKET



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Primary Users

- Hospitals, clinics, and individual healthcare practitioners
- Government and Regulatory Bodies, Health-Tech Startups and Medical Device Manufacturers

Secondary Stakeholders

- Health insurers, pharmaceutical companies, and research organizations
- Blockchain-based health insurance platforms for claim verification
- Universities conducting blockchain research for healthcare applications.

Market Size

- the healthcare blockchain market was valued between \$7 billion to \$8 billion in 2023
- Towards Healthcare predicted: The market size reached USD 13.7 billion in 2024 and is projected to hit around USD 25.52 billion by 2034

Initial Focus

- Regions with advanced digital health infrastructures (e.g., US, EU)
- Decentralized Identity Management is crucial for patients and insurers.
- Supply Chain Management is crucial for Medical Device companies



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FUTURE ENHANCEMENTS

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Selective attribute disclosure

- Patients can demonstrate proof of specific medical conditions (vaccination, for instance) without disclosing all records

Implementing Zero-Knowledge Proofs (ZKPs)

- Privacy-Preserving Authentication – Proving an identity without revealing personal data
- Scalability & Efficiency – Use zk-SNARKs/zk-STARKs to optimize blockchain transactions

Credential issuance and verification

- Authenticate medical qualifications without exposing extra details
- Reduces fraud in medical licensing and credentialing

Credential issuance and verification

- Use fingerprints, facial recognition, or iris scans for authentication
- Prevent identity theft and unauthorized access in healthcare systems
- Privacy & Compliance – Ensure secure biometric storage and processing while meeting GDPR/HIPAA standards.

Recovery Mechanisms

- Use social recovery or multi-signature wallets to restore access (Decentralized key recovery)
- Blockchain Anchoring – Store encrypted recovery data on-chain for tamper-proof restoration.



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TEAM DETAILS

ROLE	NAME	STREAM/DEPT	YEAR	COLLEGE
Frontend Developer	Dev Patel	Mathematics and computing	1st Year	IIT Kharagpur
Frontend and Blockchain Developer	Neel Gupta	Biosciences and Biotechnology	1st Year	IIT Kharagpur
Backend and Blockchain Developer	Harshit Panthi	Mathematics and computing	1st Year	IIT Kharagpur
System Architect	Dhruv Bansal	Electrical Engineering	1st Year	IIT Kharagpur



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